

Project Summary

Date	12 July 2026
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns

Project Summary

The **Plugging into the Future: An Exploration of Electricity Consumption Patterns** project was developed to analyze electricity consumption across different Indian states and regions using modern data analytics techniques. The project transforms raw electricity consumption data into meaningful insights through structured database management, interactive dashboards, and a web-based visualization platform.

The dataset was collected in CSV format, cleaned, and processed using Microsoft SQL Server. SQL queries were used to perform data extraction, filtering, aggregation, and trend analysis. The processed data was connected to Tableau Public to create interactive dashboards and stories that present state-wise, region-wise, monthly, quarterly, and yearly electricity consumption patterns. A Flask web application was developed to integrate these dashboards into a responsive interface built with HTML, CSS, and Bootstrap.

Objectives Achieved

- Collected and prepared the electricity consumption dataset.
- Designed and implemented a Microsoft SQL Server database.
- Performed SQL-based data analysis.
- Developed interactive Tableau dashboards and stories.
- Built a Flask web application for visualization.
- Published dashboards using Tableau Public.
- Maintained project documentation and GitHub repository.

Technologies Used

- Python
- Flask
- Microsoft SQL Server
- Tableau Public
- HTML
- CSS
- Bootstrap
- Git & GitHub

Key Outcomes

- Interactive visualization of electricity consumption trends.
- State-wise and region-wise comparative analysis.
- Monthly and quarterly consumption insights.
- Analysis of electricity usage before and after the COVID-19 lockdown.
- Responsive web interface for accessing dashboards and reports.

Future Enhancements

- Integrate real-time electricity consumption data.
- Add predictive analytics for future demand forecasting.
- Develop user authentication and role-based access.
- Optimize database performance for larger datasets.
- Deploy the application to a cloud platform for public access.

Conclusion

The project successfully demonstrates how data analytics can be used to understand electricity consumption patterns and support data-driven decision-making. By combining Microsoft SQL Server, Tableau Public, Flask, and web technologies, the solution provides an efficient, interactive, and scalable platform for analyzing electricity consumption data. The project meets its objectives and can be further extended with advanced analytics and real-time data integration in the future.